Princples of Observability

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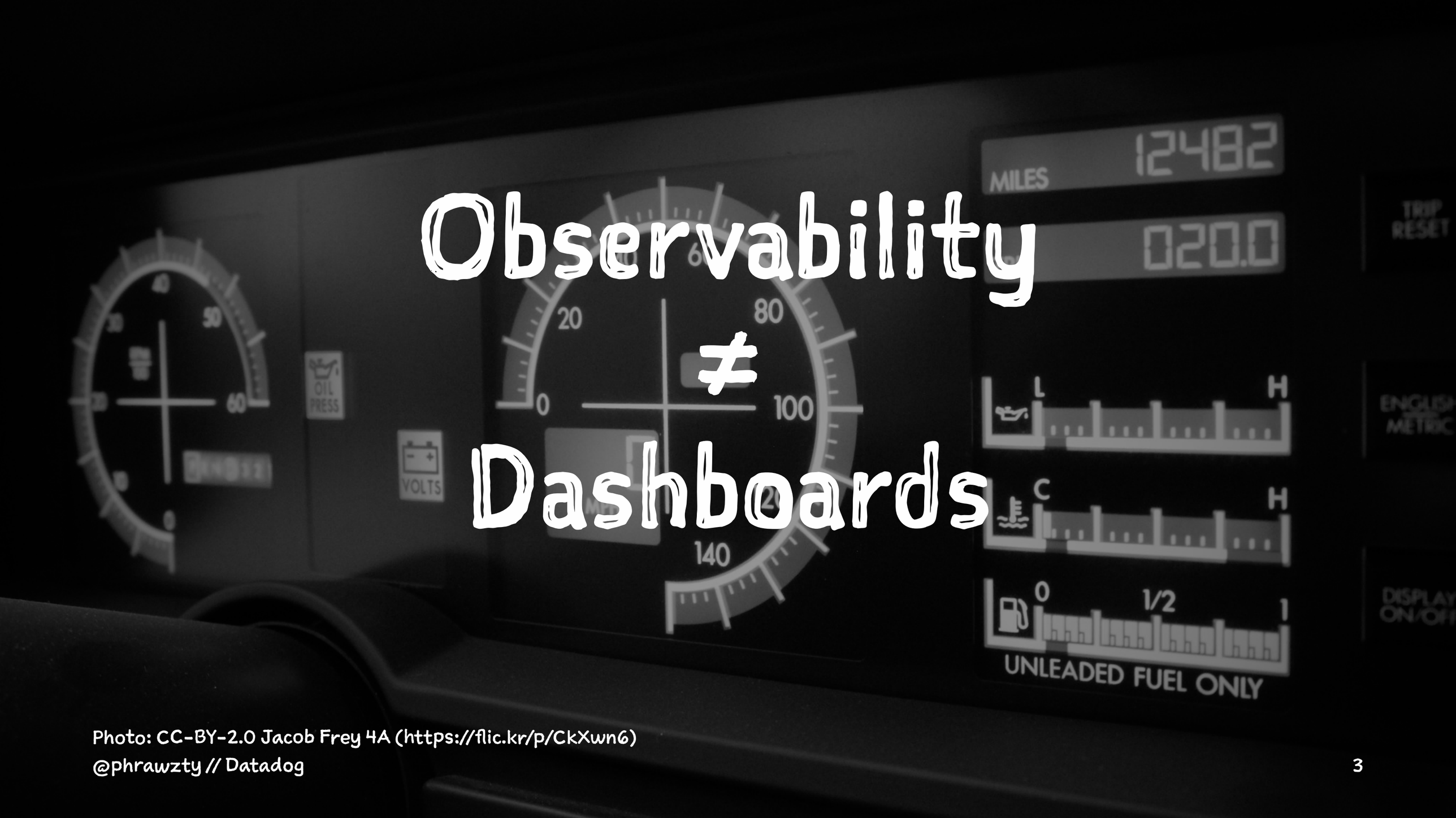
👾 Community Team

Photo: CC-BY-2.0 Steve Parker (https://flic.kr/p/afiseM)



Observability ≠ Dashboards

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Observability ≠ DevOps

Photo: © DevOpsDays Zurich (https://flic.kr/p/TNbJPm)

A brief history

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Observability is a measure of how well internal states of a system can be inferred from knowledge of its external outputs.

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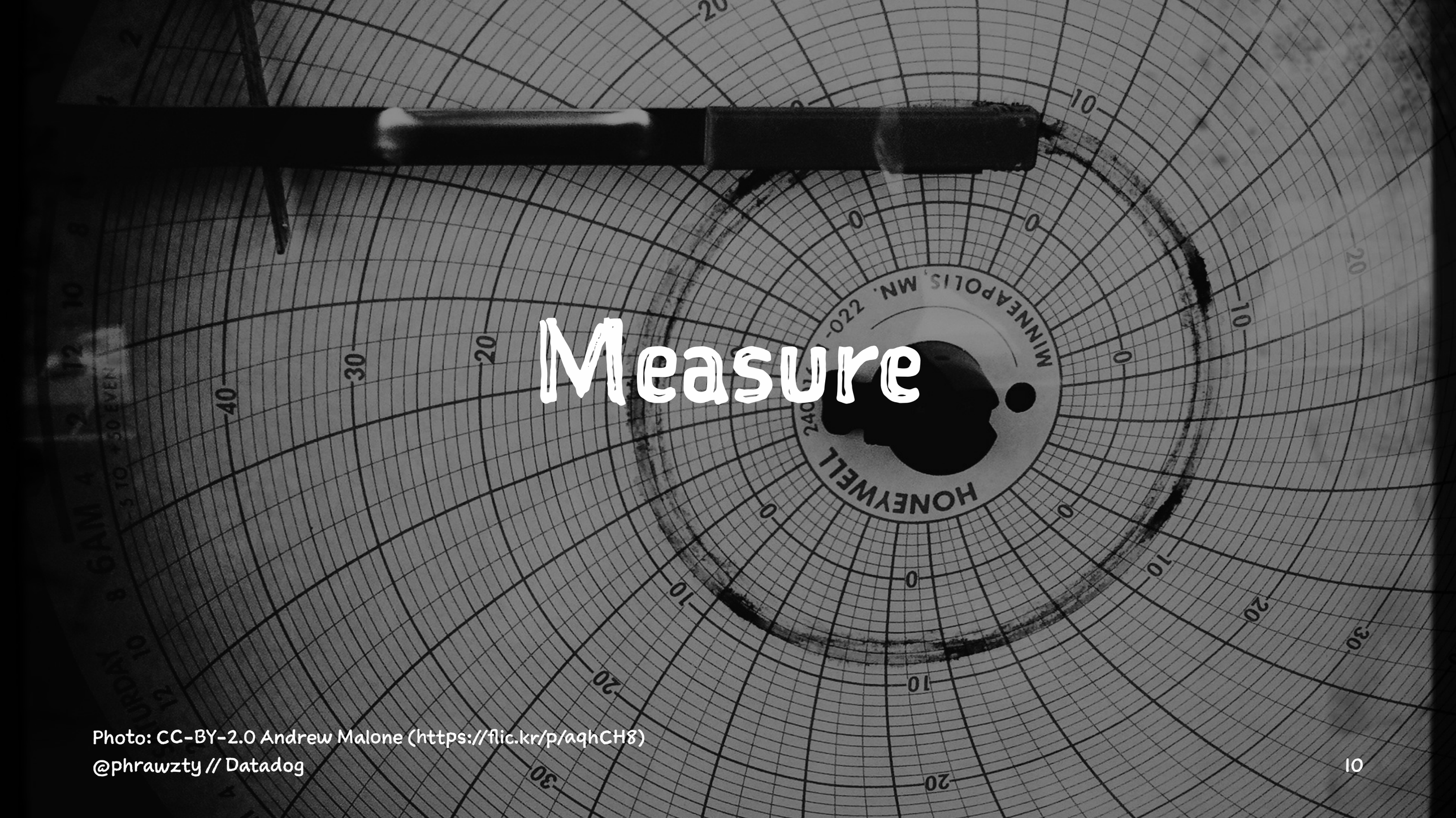

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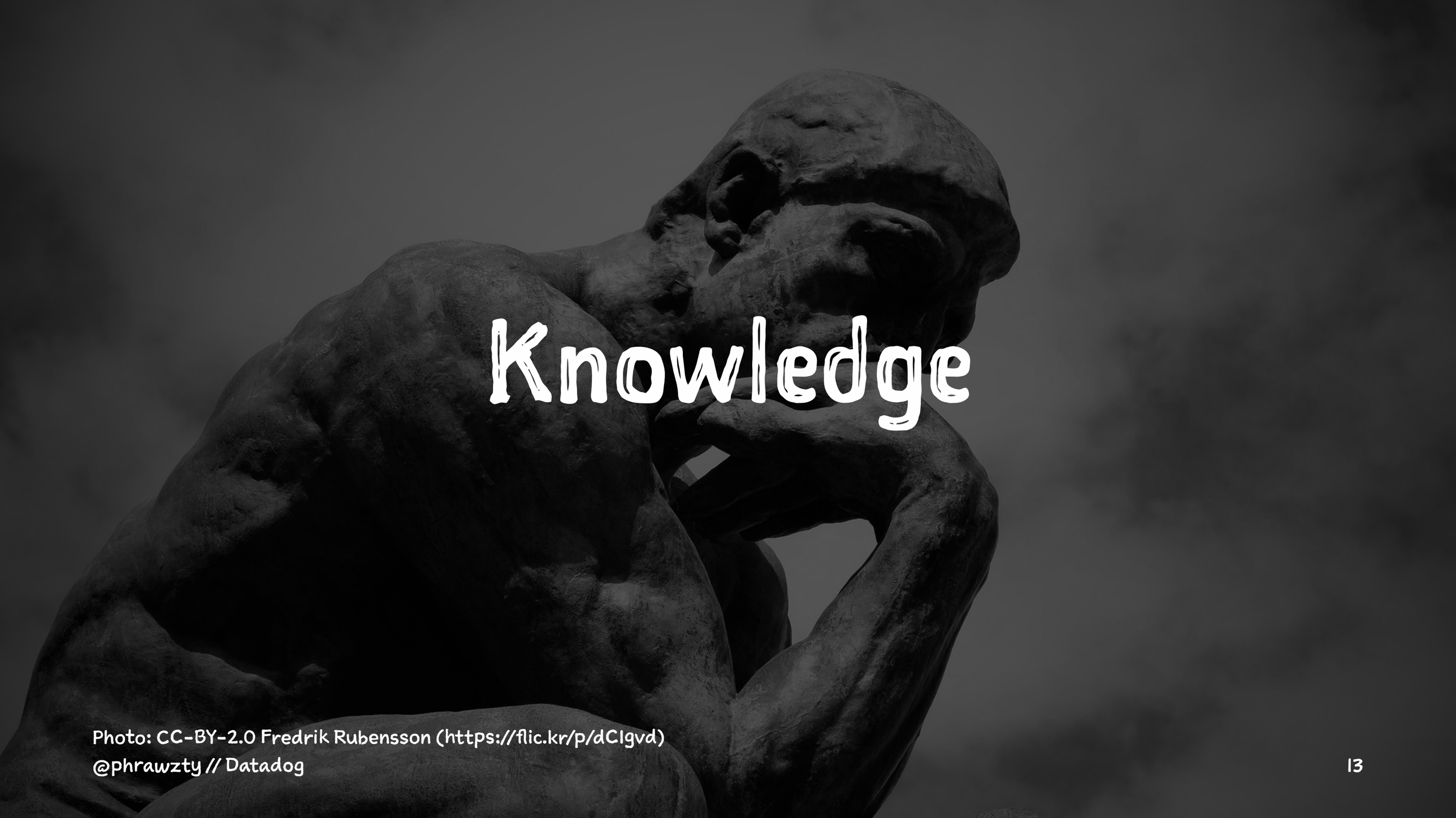

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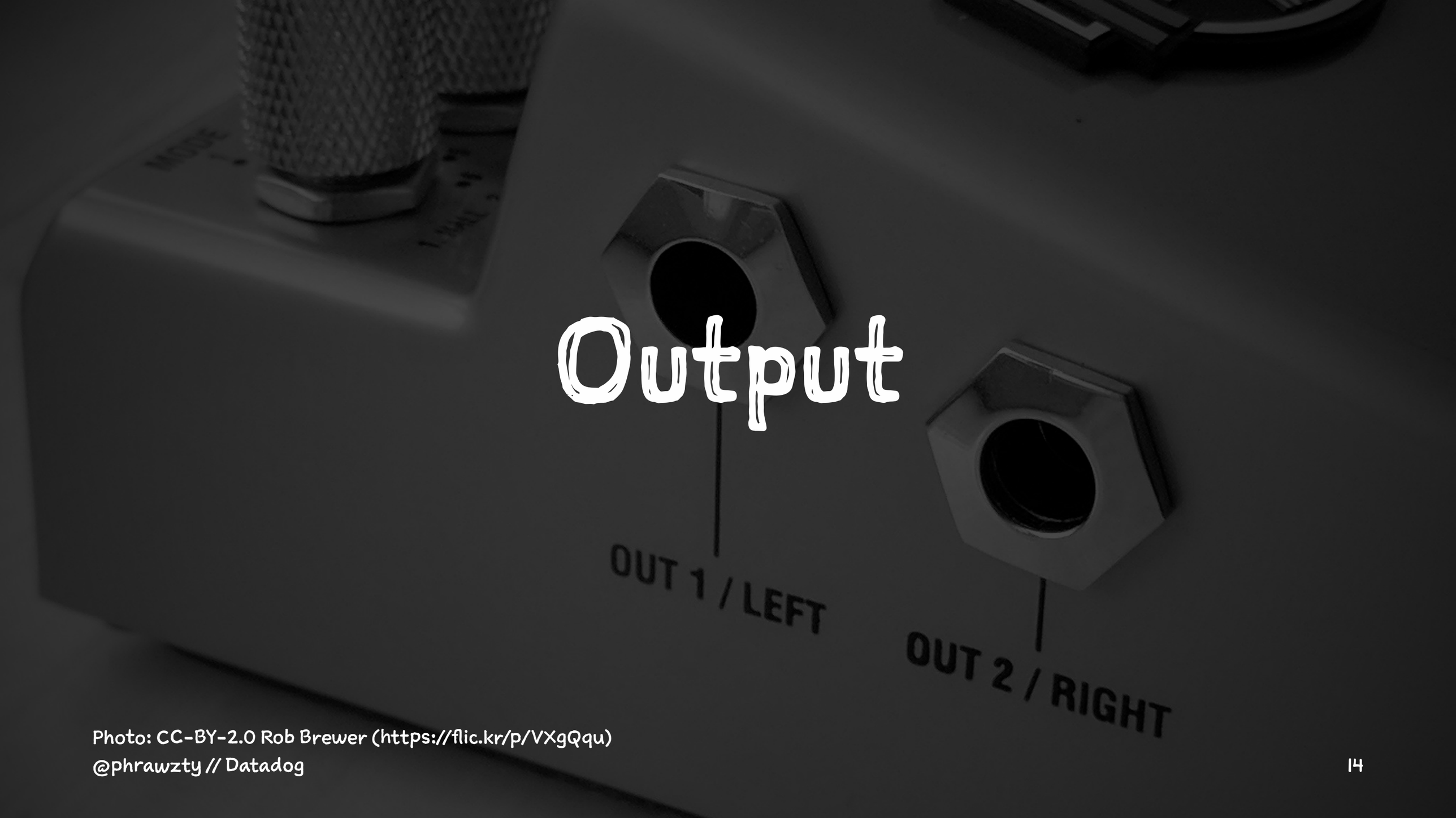

Output

Photo: CC-BY-2.0 Rob Brewer (https://flic.kr/p/VXgQqu)

Context is important

(...and control theory is vast)

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State observer

Photo: CC-BY-SA John Dyhouse (https://flic.kr/p/7ErCMc)

Multi & Bounding observer

Interval & Luenberger observers

Positive systems! Moving horizon estimations!

Kalman filters in general topological spaces!

Photo: CC-BY-SA https://en.wikipedia.org/wiki/State_observer

Photo: CC-BY-2.0 Andrew (https://flic.kr/p/xy6rdF)

Events, Signals, Telemetry, Paths, Stories...

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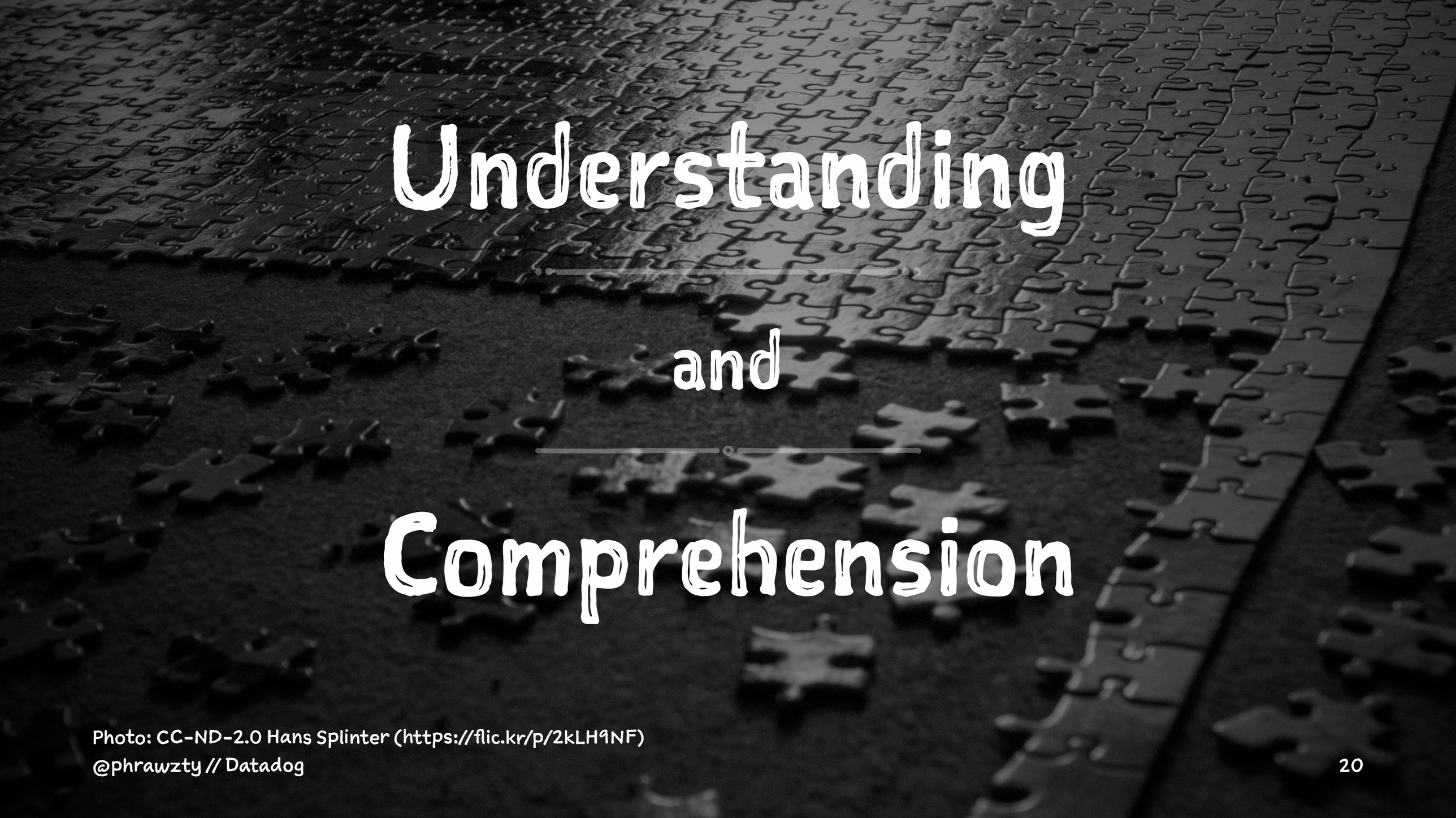

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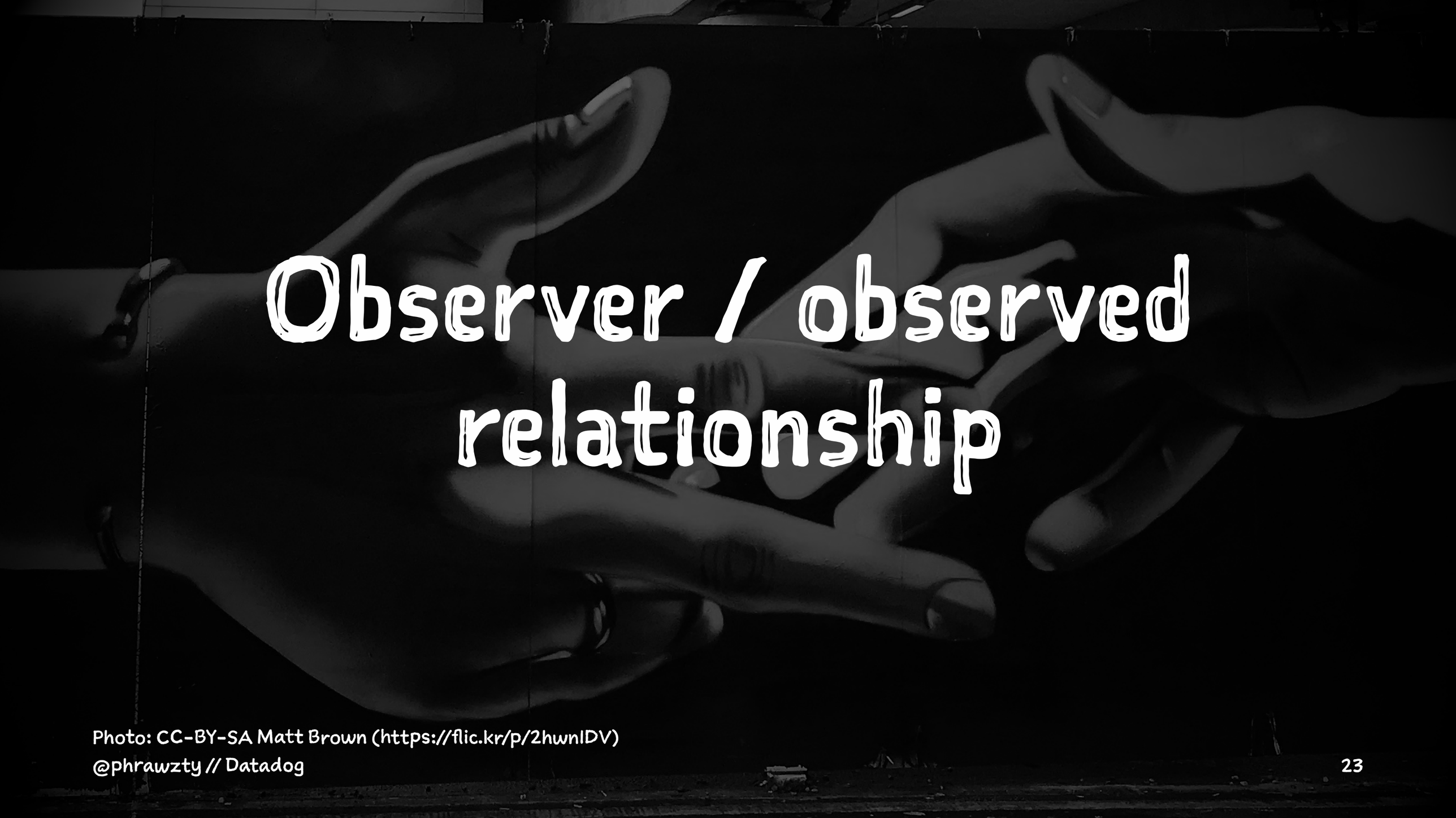


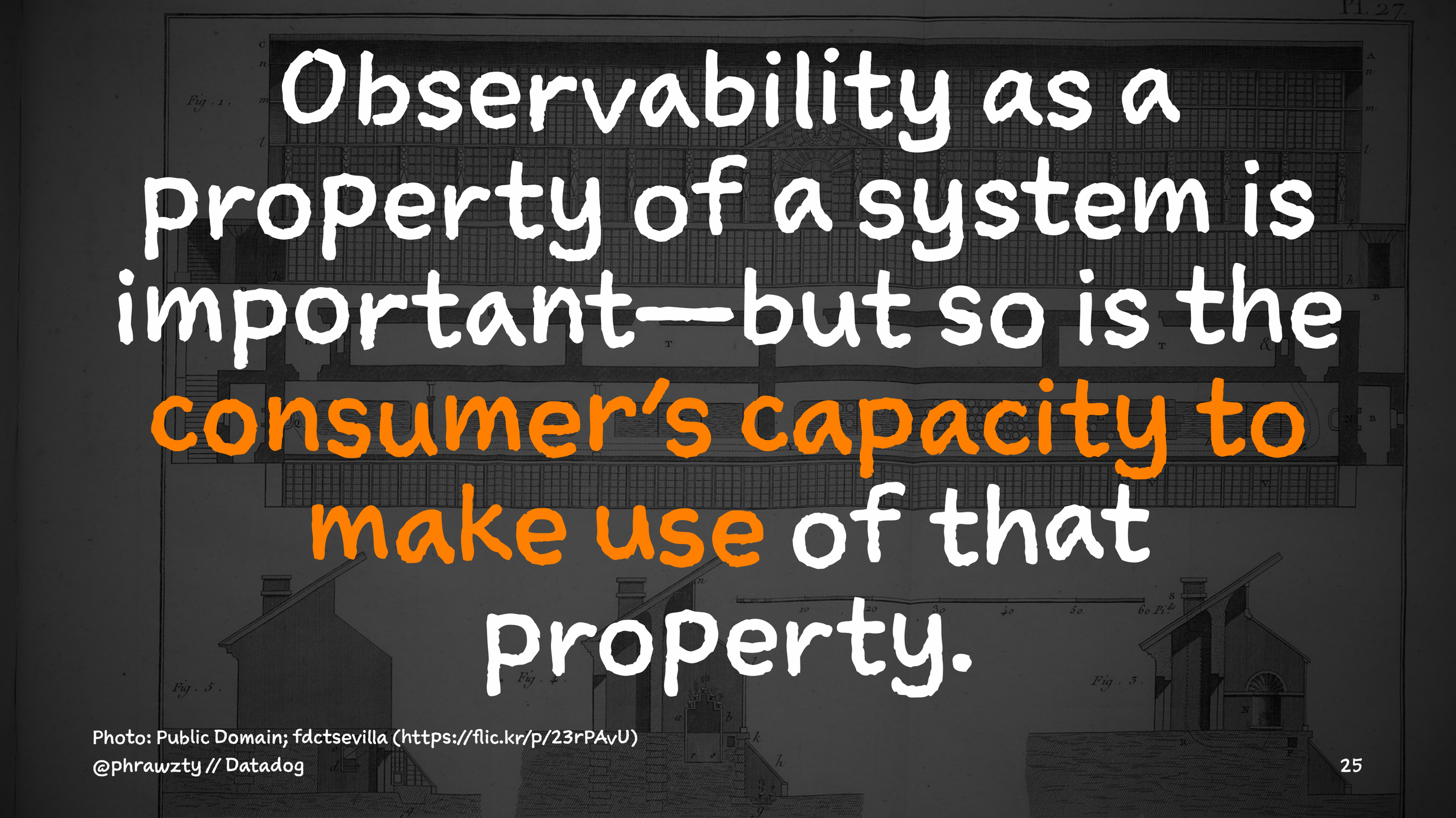
Observer / observed relationship

Photo: CC-BY-SA Matt Brown (https://flic.kr/p/2hwnIDV)

You are more

than a state observer

Photo: Public Domain; Nesster (https://flic.kr/p/ob5uhX)



Observability as a property of a system is important—but so is the consumer's capacity to make use of that property.